

Nysa Sinha

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EDUCATION

Vellore Institute of Technology, Vellore | B.Tech, Computer Science (IoT) | CGPA: 8.00 Expected 2027
Relevant coursework: Data Structures & Algorithms, DBMS, Software Engineering, Operating Systems, Object-Oriented Programming, Web Development

EXPERIENCE

Software Development Intern | Zeta, Bangalore May 2026 – Jun 2026

- Built an automated testing framework in Python with Selenium / Playwright that runs tests, records each run and generates detailed reports for a production finance application, cutting test execution time by **~75%** versus manual testing.
- Engineered a UI automation suite of **150+ test cases** covering critical user flows, improving release traceability and catching regressions before deployment.

Development Intern | Pelorus Technologies, Mumbai May 2025 – Jul 2025

- Designed secure login and authentication workflows and implemented role-based access control (RBAC) so that admins and regular users could only access the features and data permitted for their role.
- Applied credential protection and secure access control best practices to harden platform access and prevent unauthorized use.

Design Intern | Aptifyr Oct 2024 – Nov 2024

- Designed UI components and branding materials for the company's digital platforms.

PROJECTS

Unsaid — Anonymous Peer-Support Platform | unsaid-opal-chi.vercel.app

Next.js 16, React 19, TypeScript, Tailwind CSS, Supabase (PostgreSQL, Auth), Vercel

- Built a full-stack anonymous peer-support platform connecting askers with vetted peer “Repliers” across **8 life domains**, with domain-specific assessments (**40 scenario-based questions**), moderator-approved “Qualified” badges and preference-based question-replier matching.
- Enforced anonymity at the database layer using **PostgreSQL Row-Level Security** and identity-safe views, so an asker's identity can't be retrieved even through direct API calls.
- Implemented trust and safety features: server-side personal-information detection, per-user rate limiting, auto-hiding content after 3 community reports, crisis-language detection with helpline resources, and an admin moderation dashboard.

AI Fashion Assistant | Flutter, FastAPI, Python, PostgreSQL, Supabase

- Built an AI-powered personalised outfit recommendation system driven by body type, skin tone, occasion, weather and style preferences.
- Developed the Flutter mobile app and a FastAPI REST backend with PostgreSQL (Supabase) for user profiles and preferences.

Industrial Gas Leak Detection System | ESP32, MQ2, MQ7, DHT11, Python, Computer Vision Central Hackathon Finalist

- Built an IoT industrial safety system integrating gas, temperature, humidity and occupancy sensors for continuous environmental monitoring.
- Combined sensor data with AI-based camera monitoring to detect hazardous conditions and trigger real-time alerts for rapid emergency response.

Real-Time Traffic Management System | Python, YOLO, OpenCV, ESP32, Fog Computing

- Integrated YOLO with OpenCV to detect and count vehicles from live camera feeds for traffic density estimation.
- Used fog computing to process data closer to the edge, reducing latency for faster real-time traffic decisions.

TECHNICAL SKILLS

Languages: Java, Python, SQL, TypeScript, JavaScript, C++, C

Frontend & Mobile: React, Next.js, Tailwind CSS, Framer Motion, Flutter

Security: Authentication, RBAC, PostgreSQL Row-Level Security, credential protection

Tools & Concepts: Git, GitHub, Vercel, DSA, OOP, DBMS

Backend & Databases: PostgreSQL, Supabase, FastAPI, REST APIs

Testing: Automated test frameworks, UI test automation, Selenium / Playwright

AI/ML & IoT: YOLO, OpenCV, ESP32, Arduino, fog computing

Languages Spoken: English, Hindi, Marathi, Spanish (elementary)

ACHIEVEMENTS

- Smart India Hackathon — cleared Stages 1–4
- Central Hackathon — Finalist out of 150+ teams
- 3rd Place, Add-a-thon — GraVITas '23, VIT